Why to have a very high volt. for transmission purpose?

PageWork 17/01/2023

Tuesday

EE0005 (Electrical Machines & lab)

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(15 credit) → 3T+1T
→ 3L

* Monday (2:00pm – 4:30 pm)

Mon – 9:00am
Tue – 8:00am
Wed – 1:00 pm
Fri – 11:00am → Tut.

(T) Q1 – 15m
Q2 – 15m
Tut. – 5m → Marks for submission in time
End exam – 65m
100m.

(L) Cont. eval. (exps.) – 40m
Exam – 10m
50m

* What shall we study?

→ Transformers ‖ Electro magnetic system
→ Generators ‖ Electro-mechanical energy conversion systems
AC/DC → Motors
→ Magnetic ckts

⊙ Constructional details
⊙ Working principle
⊙ Develop models
⊙ Analyze
⊙ Characteristics

BOOKS:
Electric Machines – Kothari & Nagrath
Electric Machinery – Fitzgerald

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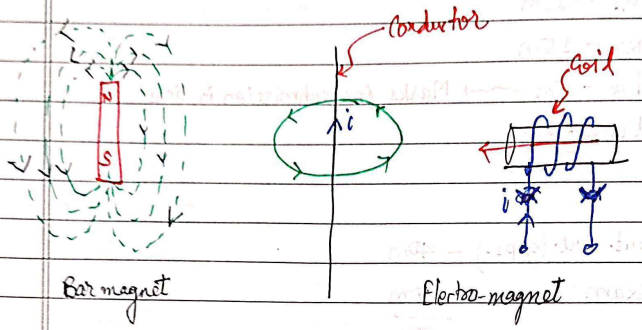
Learning Outcomes: The students are expected to
a) Understand & differentiate b/w different varieties.
b) Understand the basic nameplate specifications
c) Analyze and quantify the performance metrics
d) Understand the principles of control & operation.



* Christian Oersted (1820) → Magnetic field produced by current carrying conductor.

* Andre Marie Ampere (1820) → Force between current carrying conductors

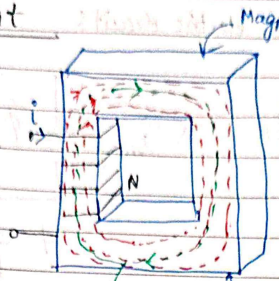
* Michael Faraday (1831) → Law of electromagnetic induction

* Lenz (1834) → Polarity of the induced emf & the resulting current

* Maxwell (1860) → Relationship between current & magnetic field

Unit of mmf : ampere-turn (AT)

Magnetic Circuit



High permeability require to confine flux (Ideally we will need zero leakage of flux)

$Ni = \oint \vec{H} \cdot d\vec{l}$ — mmf — magnetomotive force

$Hm lm$

$\therefore Ni = Hm lm$

$\Rightarrow Hm = \frac{Ni}{lm}$

$\because \phi = B \cdot A$ & $B = \mu H$

$\Rightarrow H = \frac{Ni}{l}$

$\frac{B}{\mu} = \frac{Ni}{l}$

(or) $\frac{\phi}{A\mu} = \frac{Ni}{l} \Rightarrow \boxed{\phi = \frac{Ni}{(l/\mu A)}}$

$\left(k = \frac{l}{\mu A}\right)$ Reluctance

$\Rightarrow \phi = \frac{Ni}{(lA/\mu)}$

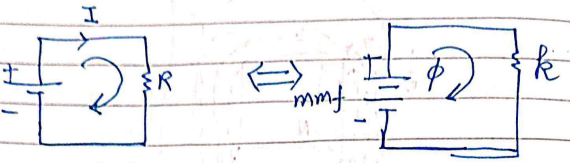
$\mu = \mu_0 \mu_r$

Thus, $\phi = \frac{mmf}{k}$

$\rightarrow \mu_r \uparrow, \phi \uparrow$

$\rightarrow k \downarrow, \phi \uparrow$

Equivalence with electric circuit:



$I = \frac{Emf}{R}$ $\qquad \phi = \frac{mmf}{k}$

↳ In both above ckts, all corresponding parameters are same except R & k, where R depicts lossy behaviour of ckt but k doesn't.

Monday

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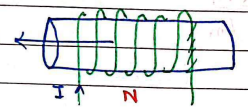
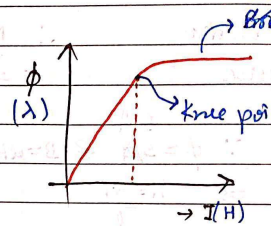
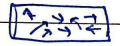
Magnetization:





Flux(ϕ)

Flux linkage(λ)

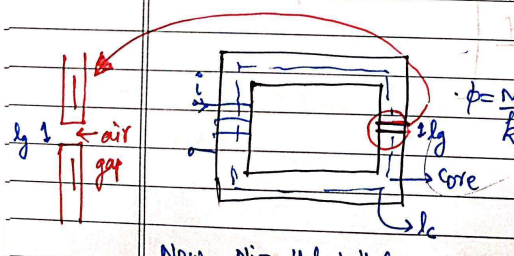


$H = \frac{NI}{l} \Rightarrow I = \frac{Hl}{N}$

$\lambda = LI$

$L = \frac{\lambda}{I} = \frac{N\phi}{I} = \frac{N(B \cdot a)}{Hl/N} = \frac{N^2(H\mu)a}{\mu l}$

$\Rightarrow \boxed{L = \frac{N^2}{(l/\mu a)} = \frac{N^2}{k}}$ (unit: Henry)



Now, $Ni = H_c l_c + H_g l_g$

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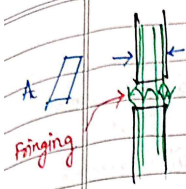


Ideally, we should assume no flux leakage through air gap.

$\therefore \boxed{\phi_c = \phi_g}$

Now, $Ni = \frac{B_c l_c}{\mu_0 \mu_r} + \frac{B_g l_g}{\mu_0} = \frac{\phi_c l_c}{a \mu_0 \mu_r} + \frac{\phi_g l_g}{a_g \mu_0}$

$\Rightarrow Ni = \phi \left[k_c + k_g \right]$ $\quad k_c = \frac{l_c}{a \mu_0 \mu_r}$, $k_g = \frac{l_g}{a_g \mu_0}$

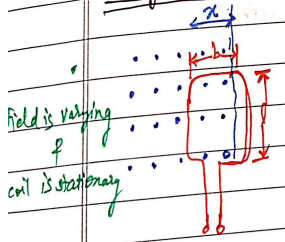
$\Rightarrow \boxed{\phi = \frac{Ni}{k_c + k_g}}$ (or) $\phi = \frac{Ni}{k_g \left(1 + \frac{k_c}{k_g}\right)}$

$\because \frac{k_c}{k_g} << 1$;

$\boxed{\phi \approx \frac{Ni}{k_g}}$

- field/flux (core)
- coil/conductor

* Faraday's law of EM induction:



$B = B_m \sin(\omega t)$ — ①

$e = \frac{d(\lambda)}{dt} = \frac{d(N\phi)}{dt} = N \frac{d\phi}{dt} = N \frac{d(Ba)}{dt}$

$\Rightarrow e = N \frac{d}{dt}(B_m \sin(\omega t) a)$

$= N B_m a \omega \cos(\omega t)$

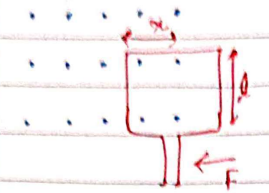
$\Rightarrow \boxed{e = B_m \cdot a \cdot \omega \sin(\omega t + \pi/2)}$ — ② (static induced emf)

$V - iR - N^2 \frac{di}{dt} = 0$

$V = \sin \omega t = iR + N^2 \frac{di}{dt}$

$\phi = \frac{mmf}{\mathcal{R}} = \frac{Ni}{\mathcal{R}}$

$e = N \frac{d\phi}{dt} = \frac{N^2}{\mathcal{R}} \frac{di}{dt} = iR$



B → constant

$e = \frac{d}{dt}(B \cdot a)$

$= B \frac{dq}{dt}$

$= B l \frac{dx}{dt}$

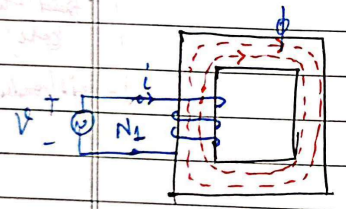
Field mag. is const but coil is moving

$\Rightarrow e = Blv$; (Dynamically induced emf or Speed emf)

~~Field & coil is rest~~

- Field is varying & coil is moving

$e = \frac{d}{dt}\left[(B_m \sin \omega t) l x \right]$



$\phi = \phi_m \sin(\omega t)$ — ①

$e_1 = N_1 \frac{d\phi}{dt}$

$= N_1 \phi_m \omega \cos \omega t$

$= \phi_m N_1 (2\pi f) \cos \omega t$

$\left[\frac{V_0 \sin\omega t}{N_2} = k\left(\frac{iR}{N^2} + \frac{N\frac{d\phi}{dt}}{N^2}\right)\right] e^{\frac{kRt}{N^2}} \Rightarrow \frac{kV_0 \sin\omega t \cdot e^{\frac{kRt}{N^2}}}{N^2} = \frac{d}{dt}\left(i e^{\frac{kRt}{N^2}}\right)$

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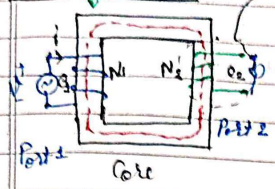
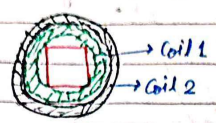
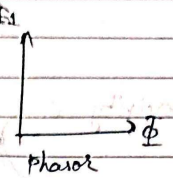
Tuesday

TRANSFORMER

→ Coil 2 is wound on coil 1



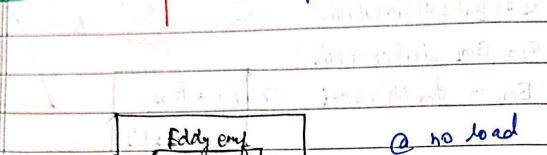




$\phi = [\phi_m] \sin\omega t$ — ①

$e_1 = [\phi_m N_1 \omega] \sin(\omega t + \pi/2)$ — ②

Not should be beyond knee point



$e_1 = E_{m1} \sin(\omega t + \pi/2)$

$E_{m1} = \phi_m N_1 \times 2\pi f \Rightarrow \boxed{\phi_m = \frac{E_{m1}}{2\pi f_1 N_1}}$

$E_{m1} = V_1$

For ideal behaviour, and for const. N_1,

$\phi_m \propto \left(\frac{V_1}{f_1}\right)$ → V/f control

$e_2 = E_2 \sin(\omega t + \pi/2)$

$\phi = \phi_m N_2$

$E_2 = \phi_m N_2 \times 2\pi f$ — ③

∴ from ②, ③

True for ideal transformer ← $\boxed{\frac{E_1}{E_2} = \frac{N_1}{N_2}}$

Port 1 side ≡ Primary winding

Port 2 side ≡ Secondary winding

$\hat{e_1} = 2\pi f \phi_m N_1$

Rms value of $\hat{e_1}$, $E_1 = \frac{2\pi f}{\sqrt{2}} \times \phi_m N_1$

I_μ : Magnetising current
↳ Min current needed to magnetise the core.

$E_1 =$

$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

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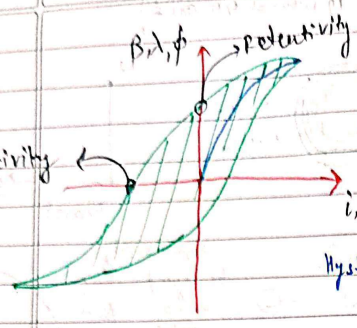
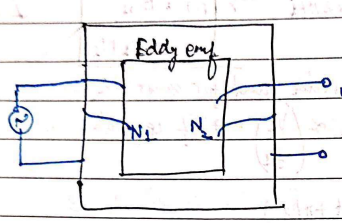


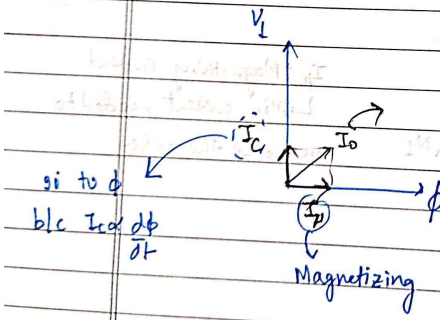
Hysteresis loss = Area in Green



I_μ and Eddy Losses:

→ Eddy current losses ∥ I_c

→ Hysteresis losses



Min current

I_o — no load input current

$P_{av} = V_1 I_o \cos\phi_o$

P, Q, S

P: Average Power
Q: Reactive Power
S: Apparent Power

i_c is to ϕ b/c $I_c \propto \frac{d\phi}{dt}$

Wednesday

Non ideality → Windings → Resistance, Flux leakage → Hysteresis
Non ideality → Core → Magnetisation, Losses → Hysteresis, Eddy current

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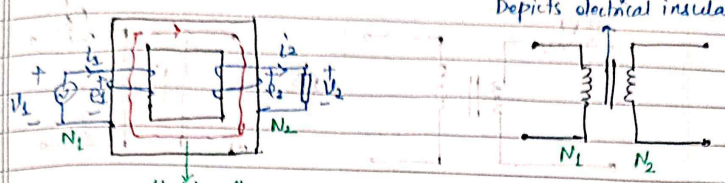
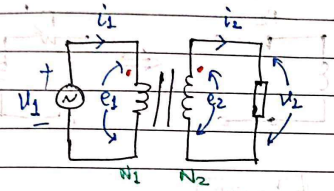
Depicts electrical insulation



Due to load current (i_2), primary winding will try to compensate the opposite flux induced and thus draws more current called "load equivalent current" from source.



Non ideality of the windings:

$$R_1 = \frac{\rho_1 l_1}{a_1}$$

$$R_2 = \frac{\rho_2 l_2}{a_2}$$

Voltage Transformation ratio (a)

$$N_1 \cdot I_1 = N_2 I_2$$

$$\Rightarrow \boxed{\frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1} = a}$$

↳ Step down T/F ; $a > 1$
Step up T/F ; $a < 1$
1:1 T/F ; $a = 1$

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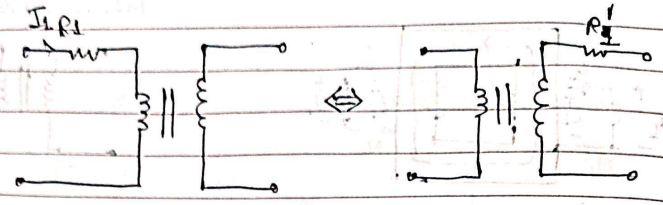
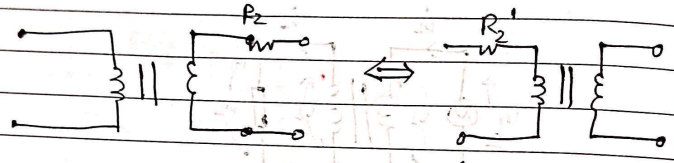


$$I_1^2 R_1 = I_2^2 R_1'$$

$$\Rightarrow \boxed{R_1' = \frac{R_1}{a^2}}$$



Similarly, $\boxed{R_2' = a^2 R_2}$

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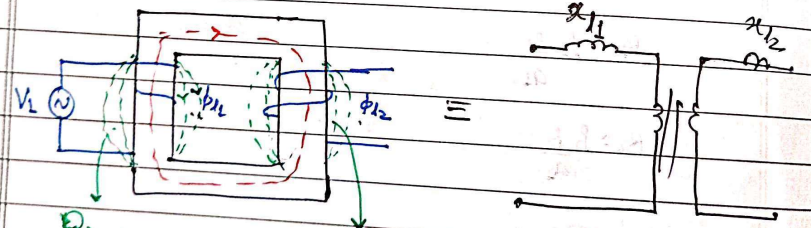

Primary leakage flux (ϕ_{l_1})

Secondary leakage flux (ϕ_{l_2})

Monday

Tuesday

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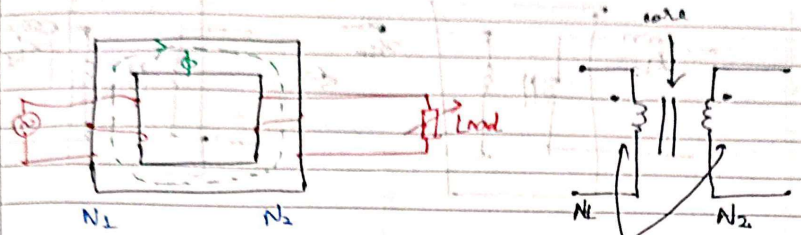
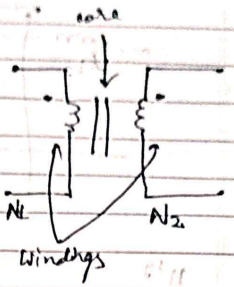
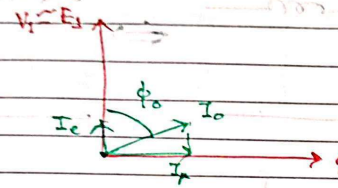
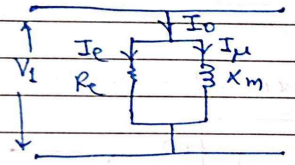




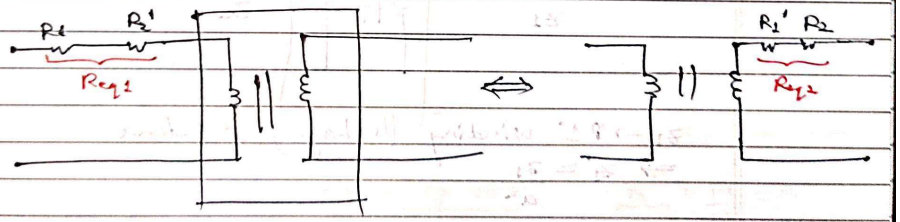


$P_0 = V_1 I_0 \cos\phi_0$

$P_i R_i S$

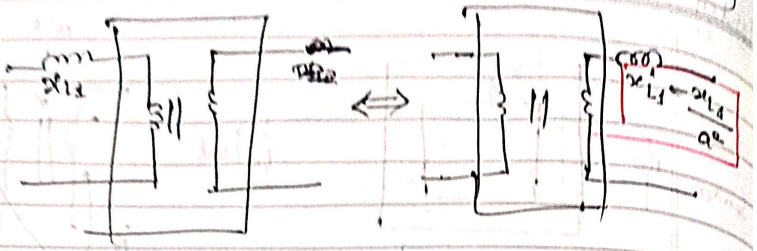


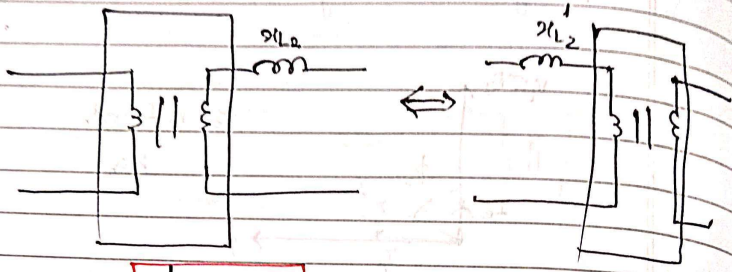
$\overline{I_0} = \overline{I_e} + \overline{I_M}$



$R_{e1} = R_1 + R_2'$

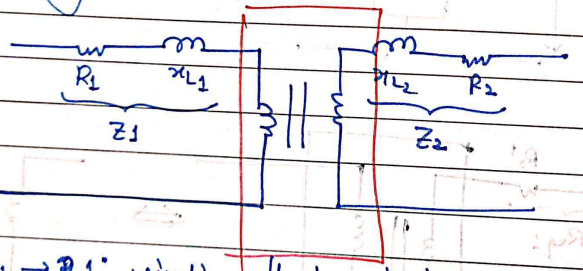
$R_{e2} = R_1' + R_2$





$x'_{L2} = a^2 x_{L2}$

∴ Combining both phenomenons!

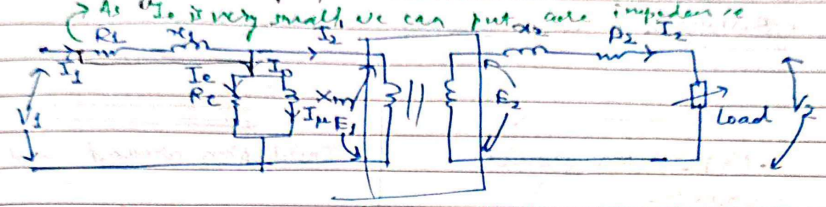


$Z_1 \rightarrow$ Primary winding (leakage) impedence

$\Rightarrow Z_1' = \frac{Z_1}{a^2}$

On combining windings & core non-idealities!

As I_0 is very small, we can put core impedance

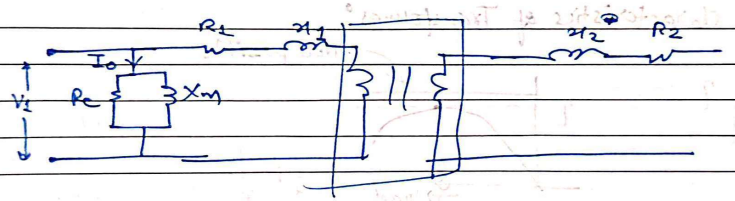


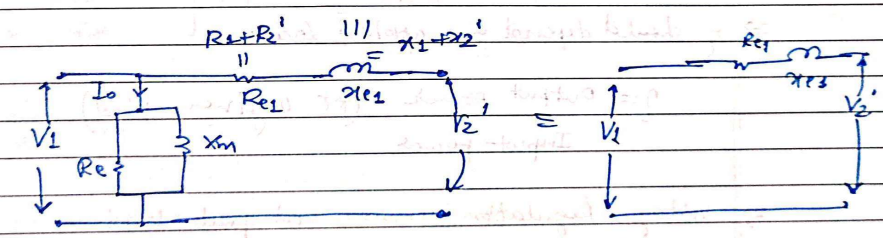
$I_0 \approx 0.5\%$ of full load current

Exact eq. ckt of the single phase Transformer.

I_2' : Equivalent current drawn from source to compensate opposite flux from 2° windings.

$\therefore \vec{I_1} = \vec{I_2'} + \vec{I_0}$





Approx. eq. ckt

$\frac{V_2'}{V_2} = \frac{N_1}{N_2} = a$

Tuesday

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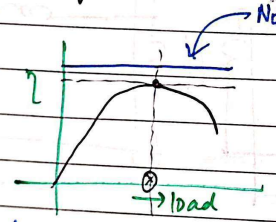
- Burning of windings!

• No insulation around conductor	• Insulation around conductor
• Conductor melts / fuse when current through is greater than rated value.	• If applied voltage is greater than rated voltage, voltage may leak from coating to adjacent turning. This will produce a high current which may lead to burn

- Characteristics of Transformer:

1) η

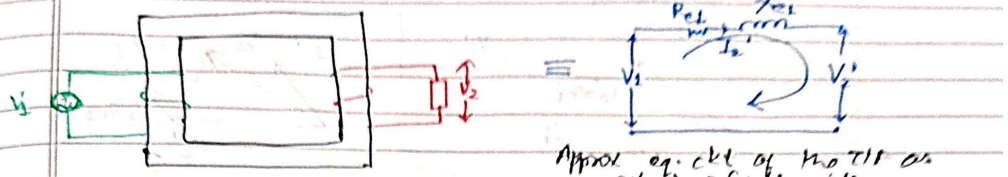


$\Rightarrow \eta$ should depend on applied load.

$\eta = \frac{\text{Output power}}{\text{Input power}}$ (for a given load)

2) Voltage Regulation: common at peak load.

Wednesday



Approx eq. ckt of 1Ø TF ref to primary side

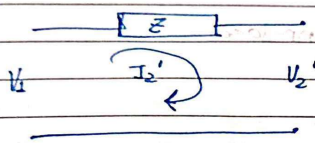
$R_{e1} = R_1 + R_2'$

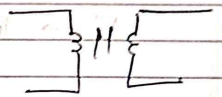
$X_{e1} = X_1 + X_2'$

① Efficiency

② Voltage Regulation

↳ Simplying further,





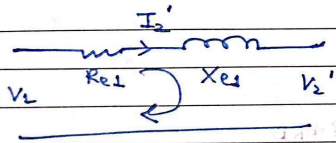
$$\frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

$$V_1 I_1 = V_2 I_2$$

TF Transfers power from 1° side to 2° side.

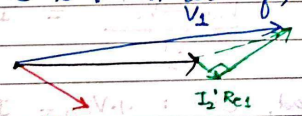
- Load can be —
  - a) Resistive
  - b) Inductive
  - c) Capacitive

} $P = VI\cos\phi$



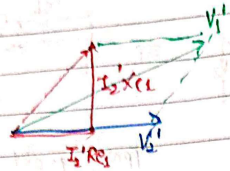
$$\bar{V}_1 = \bar{V}_2' + \bar{I}_2'(R_{e1} + jX_{e1})$$
$$= \bar{V}_2' + \bar{I}_2' R_{e1} + j\bar{I}_2' X_{e1}$$

$\because \bar{V}_2'$ is taken as ref,

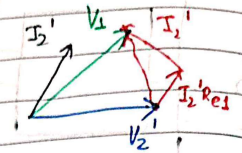


Phasor diagram of the T/F for inductive loads

For resistive load,



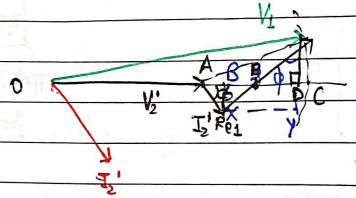
For capacitive load,



$V_{2,NL}'$ or $V_{20}' = V_1$

↳ As soon as inductive load is removed, V_2' jumps to V_1, which may causes burning of appliances, thus voltage regulation is needed.

V_2' → load voltage @ no-load



$V_1 = OC \approx OD$

$OD = OA + AD$

$V_{20}' = V_2' + AB + BD$

$= V_2' + I_2' R_{e1}\cos\phi + I_2' X_{e1}\sin\phi$

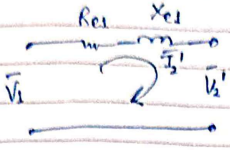
$$\frac{V_{20}' - V_2'}{V_2'} = \frac{I_2' R_{e1}\cos\phi + I_2' X_{e1}\sin\phi}{V_2'}$$

$$\therefore V_{reg} = \frac{V_{20}' - V_2'}{V_2'} = \frac{I_2' R_{e1}\cos\phi + I_2' X_{e1}\sin\phi}{V_2'}$$

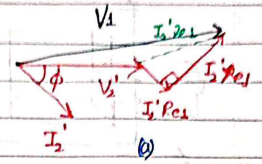
↳ For resistive load, $\phi = 0$: $\therefore V_{reg} = \frac{I_2' R_{e1}}{V_2'}$

Monday

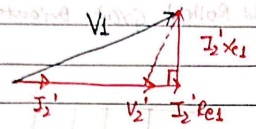
PageWork 06/02/2023

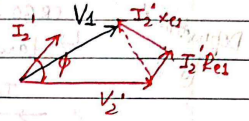


$\bar{V}_1 = \bar{V}_2' + \bar{I}_2'(R_{e1} + X_{e1})$



(a)





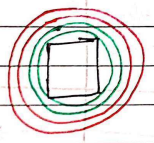
(b)

a) Inductive Loads
b) Resistive loads
c) Capacitive loads

} for same V_2' (Rated output voltage)

$$\% V_{reg} = \frac{V_{20}' - V_2'}{V_2'} \times 100\% = \frac{I_2' R_{e1}\cos\phi \pm I_2 X_{e1}\sin\phi}{V_2'}$$

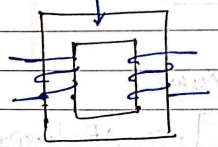
↳ + (lagging pf)
↳ - (leading pf)



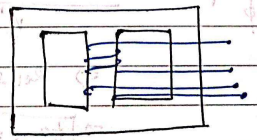
HV Windings
LV Windings

- Leakage flux (X_{e1}) can be minimised by winding 1' & 2' as close tight as possible.

Core structure



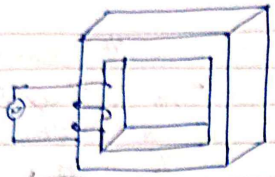
Core type
→ winding surrounding core



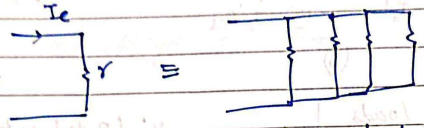
Shell type
→ Core surrounding windings
→ Lesser leakage flux.

What is the direction of eddy current?

PageWork

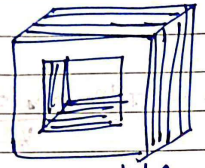


Dependents (f) → Hysteris losses ; B-H curve - loop should be small
→ Eddy current losses
CRGO steel — Cold Rolled Grain Oriented Steel
Dependents (f^2) ↳ Laminated core



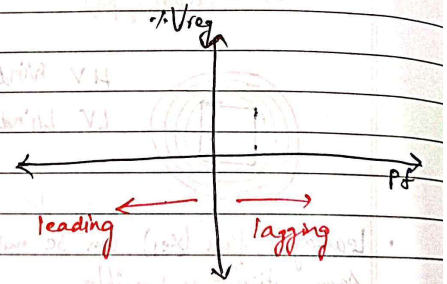
Joule heat loss reduced.

∴



Laminated Core → Reduces eddy current losses.

∴ $V_{reg} = 0$ (for leading pf)



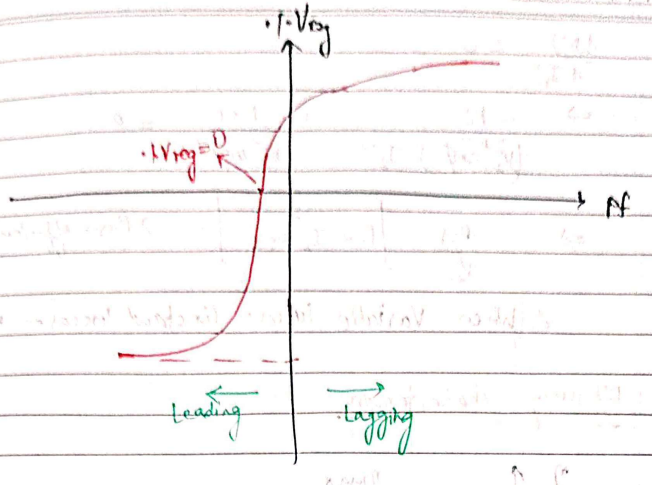
$R_{e1}\cos\phi = X_{e1}\sin\phi$

$\Rightarrow \boxed{\tan\phi = \frac{R_{e1}}{X_{e1}}}$

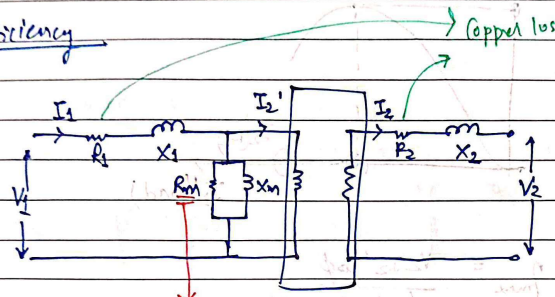
Max $\frac{d}{d\phi} V_{reg} = \frac{-I_2' R_{e1}\sin\phi \pm I_2' X_{e1}\cos\phi}{V_2'}$

$\Rightarrow R_{e1}\sin\phi = X_{e1}\cos\phi$

$\Rightarrow \boxed{\tan\phi = \frac{X_{e1}}{R_{e1}}}$ → For lagging pf.



* Efficiency



Corelosses → Don't changes with load.

$$\eta = \frac{o/p}{I/p} = \frac{V_2' I_2' \cos\phi}{V_2' I_2' \cos\phi + (P_i' + P_{cu})}$$

For given f, they are constant

∴ CONSTANT LOSSES

Iron losses

I/p = o/p + losses

Coreloss, Copper loss

Copper loss → Varies with load (More load, more loss?)

$$\therefore \eta = \frac{1}{1 + \frac{P_i' + P_{cu}}{V_2' I_2' \cos\phi}}$$

$$\Rightarrow D^r = \frac{1 + P_i'}{V_2' I_2' \cos\phi} + \frac{I_2'^2 R_{e1}}{V_2' I_2' \cos\phi}$$

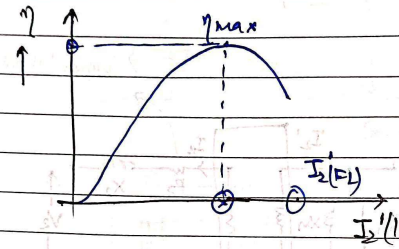
$$\frac{d(D^r)}{d I_2'} = 0$$

$$\Rightarrow -\frac{P_i'}{(V_2' \cos\phi) I_2'^2} + \frac{R_{e1}}{V_2' \cos\phi} = 0$$

$$\Rightarrow \frac{R_{e1}}{V_2'} \quad \boxed{P_i' = I_2'^2 R_{e1}} \rightsquigarrow \text{Max. efficiency.}$$

∴ When Variable losses = Constant losses → η_{max}

Efficiency characteristic



$$\therefore \eta_{max} = \frac{V_2 I_2' \cos\phi}{V_2' I_2' \cos\phi + 2 P_i'}$$

$I_2'(FL)$ → max I_2' when operated at Full load without burning

$$\because \frac{I_2'^2 R_{e1}}{I_2'^2_{(FL)}} = \frac{P_i'}{I_2'^2_{(FL)} R_{e1}} \Rightarrow \left(\frac{I_2'}{I_{2(FL)}'}\right)^2 = \frac{P_i'}{I_{2FL}'^2 R_{e1}}$$

P_i' → Core losses

$I_{2FL}'^2 R_{e1}$ → Full-load Cu losses

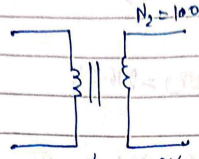
$$\boxed{k = \sqrt{\frac{P_i'}{P_{cu(FL)}}}} \rightsquigarrow \text{Fraction of } I_{2(FL)} \text{ when max. efficiency is achieved.}$$

$$\text{So, } \eta_{max} = \frac{k V_2 I_{2(FL)}' \cos\phi}{k V_2 I_{2FL}' \cos\phi + P_i' + (k I_{2FL}')^2 R_{e1}}$$

Tuesday

QUE: 11 kV
500 kVA, 11000/400 V, 50 Hz, 1-ϕ T/F, N_2 = 100 t'
Calculate approx. no. of turns of 1° winding.

ANS:



500 kVA T/F

$\frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow N_1 = \frac{11000 \times 100}{400} = 2750$ turns

QUE: (i) I_1(rated) = I_1(FL) = ? & I_2(rated) = I_2(FL) = ?
(ii) So 500 kVA = $V_1 I_1$ @ rated load or full-load.

$I_1 = \frac{500 \text{ kVA}}{11 \text{ kV}} = 45.54$ A

11/y, $I_2 = \frac{500 \text{ kVA}}{400} = 1250$ A

$e = \phi_m \omega \sin(\omega t + \phi)$

$\phi_{max} = V_1$ ⇒ Do either for 1° or 2°.

$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{\phi_m (2\pi f) N_1}{\sqrt{2}}$

$11000 = \phi_m (2\pi) f N_1$

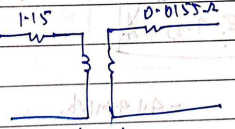
$\Rightarrow \phi_m = \frac{11000}{2\pi \times 50 \times 2750} = 11.5$ mWb.

$\phi = \phi_m \sin(\omega t + \phi)$

$V_1 = \phi_m \omega \cos(\omega t + \phi)$

QUE: 2° A 40 kVA, 2000/250 V, 1 ϕ T/F
$R_1 = 1.5\,\Omega$, $R_2 = 0.0155\,\Omega$
(i) Calculate resistance of T/F w.r.t. 2°

⇒



$I_1^2 R_1 = I_2^2 R_1'$

$R_{e2} = R_2 + R_1'$

$R_1' = \left(\frac{I_1}{I_2}\right)^2 R_1 = \left(\frac{V_2}{V_1}\right)^2 R_1$

$= R_2 + \left(\frac{V_2}{V_1}\right)^2 R_1$

$= 0.0155 + \left(\frac{250}{2000}\right)^2 \times 1.5$

$= 0.03589375$

(ii) $2000 \times 10^3 = I_{1r} V_{1r} = I_{2r} V_{2r}$
Estimate full load copper losses of T/F.

⇒ Any $I_2^2 R_{e2} = P_{cu}$

$I_{2r} = \frac{40 \text{ kVA}}{2 \text{ kV}} = \frac{4}{25} \times 1000 = 160$ A

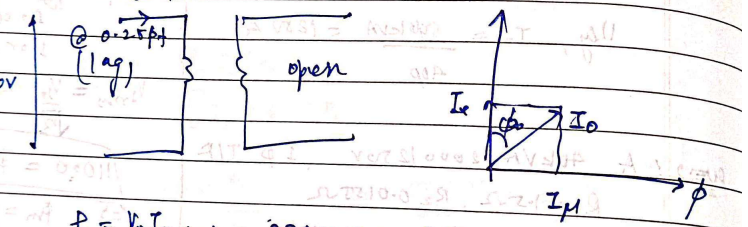
$\therefore P_{cu|FL} = 160^2 \times 0.0389 = 996.8$ W

(iii) A H-F-L., $I_2 = \frac{I_{2r}}{2} = \frac{160}{2} = 80$ A

$\therefore P_{cu} = \frac{P_{cu|FL}}{4}$

QUE: 3° A T/F: NL 4.5 A with pf = 0.25 lagging with 230 V ac main. No of turns 1° = 250. Calculate magnetizing current. Also estimate core losses in T/F.

⇒



$P_0 = V_0 I_0 \cos\phi_0 = 230 \times 4.5 \times 0.25 = 258.75$ W

$I_\mu = I_0 \sin\phi_0 = 4.357$ A

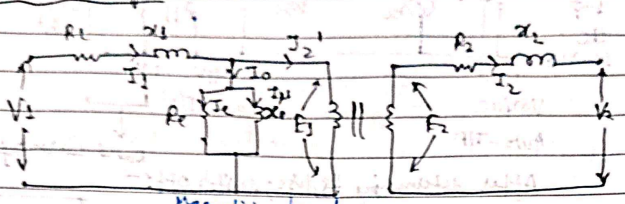
$\phi_m = ? = \frac{V_0}{4.44 \times 50 \times 250} = 4.14$ mWb.

Q. Why to open ckt HV side, no why not LV side in open ckt test?

⇒ It's it b/c to provide lower volt from volt source

↳ 'Kothari' writes its from the point of view of convenience & availability of supplying for test.

Exact model of T/F:

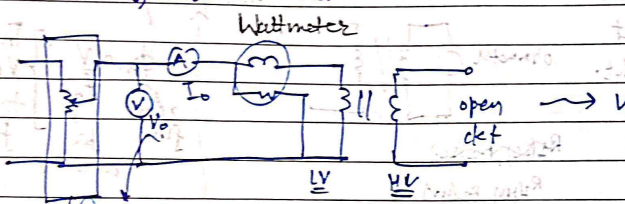


$G_c = \frac{1}{R_c}$ (℧) ; $B_m = \frac{1}{X_m}$ (℧)

In reality, we will have T/F as black box; so we need to perform a) open ckt. Test

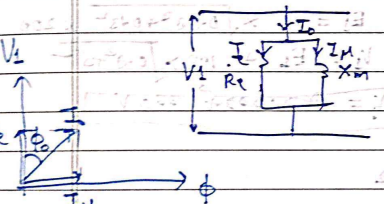
b) short ckt. Test

Open ckt Test



No load current 'Io'

Open ckt → $I_2' = 0$ ⇒ $I_1 = I_o$ ⇒ ~~Extremely~~ Very less ⇒ Cu losses ≃ 0.



$I_o = I_{oa} + I_{or}$

$I_{oa} = I_o \cos\phi_o$

$I_{or} = I_o \sin\phi_o$

Wattmeter reads Core losses.

$R_c = \frac{V_1}{I_c} = \frac{V_1}{I_o \cos\phi_o}$ can be calculated from power factor obtained

$X_m = \frac{V_1}{I_\mu} = \frac{V_1}{I_o \sin\phi_o}$ which can be obtained from wattmeter.

⇒ This test helps to find magnetizing parameters of T/F.

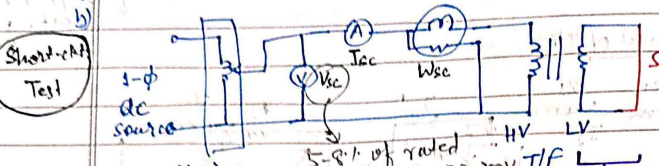
Q. Why particularly LV side is [illegible] V

⇒ Its convenient that's way. (See 'Kothari' Pg no. 82)

Wednesday

b)

Short-ckt Test



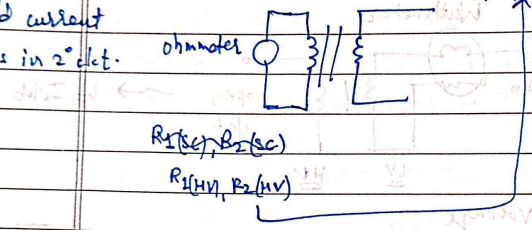
Even with small voltage at LV side, I_2 can shoot up too much. So, apply V such that only a certain rated current flows in 2° ckt.

After determining I_{sc}, W_{sc}; switch off:-

$Z_{sc} = \frac{V_{sc}}{I_{sc}}$, $R_{sc} = \frac{W_{sc}}{I_{sc}^2}$ ⇒ $X_{sc} = \sqrt{Z_{sc}^2 - R_{sc}^2}$

↳ since its difficult to obtain X_1 & X_2 separately, assume $X_{1(HV)} = X_{2(LV)} = \frac{X_{sc}}{2}$

Appropriately revert back to LV



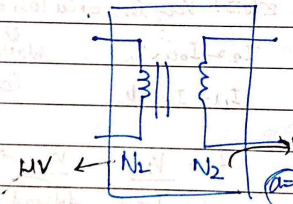
$\phi_m \propto \frac{V}{f}$

rated flux — fix at 50Hz

Then, adjust V according to get rated flux.

⇒ This test helps to find R_1, X_1, R_2 & X_2 of T/F.

GATE



$R_{HV} = 0.9\,\Omega$, $X_{HV} = 5\,\Omega$

$R_{LV} = 0.03\,\Omega$, $X_{LV} = 0.13\,\Omega$

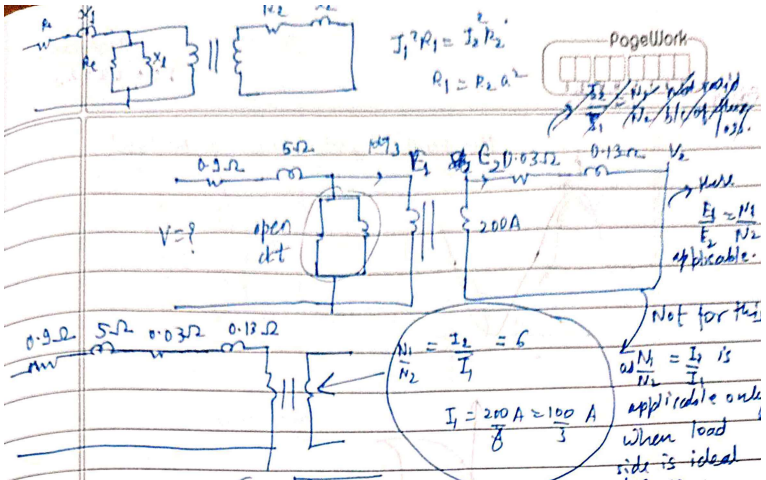
$E_2 = \sqrt{0.03^2 + 0.13^2} \times 200$

$E_1 = 6 \times \sqrt{0.03^2 + 0.13^2} \times 200$

$V_1 = E_1 + 100 \times \sqrt{0.9^2 + 5^2}$

$V_1 = \frac{1}{3}$ [illegible] 330 V

How much volt. should be applied at HV to get FL current of 200A at LV side? LV is short ckted



$I_1^2 R_1 = I_2^2 R_2$

$R_1 = R_2 a^2$

$\frac{I_2}{I_1} = \frac{N_1}{N_2}$ but not applicable.

Not for this

$\frac{N_1}{N_2} = \frac{I_2}{I_1}$ is applicable only when load side is ideal where I_2 becomes load current as well

$\frac{N_1}{N_2} = \frac{I_2}{I_1} = 6$

$I_1 = \frac{200\,A}{6} = \frac{100}{3}\,A$

$V = \frac{100}{3} \times \sqrt{0.9^2 + 5^2}$

$R_{e1} = 0.9 + 0.03 \times 6^2$

$= 1.98\,\Omega$

$X_{e1} = 5 + 0.13 \times 6^2$

$= 9.68\,\Omega$

$\Rightarrow Z_{e1} = \sqrt{R_{e1}^2 + X_{e1}^2} = 9.8\,\Omega$

$\Rightarrow V_1 = I_1 Z_{e1} = \frac{100}{3} \times 9.8 = 330\,V$

(Q.E) 150 kVA, 2400/240 V, 50 Hz, 1-ϕ TF

$R_1 = 0.2\,\Omega$, $R_2 = 2\,m\Omega$

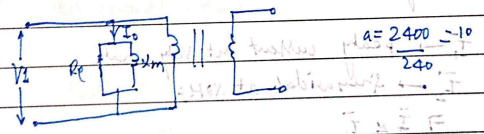
$X_1 = 0.45\,\Omega$, $X_2 = 4.5\,m\Omega$

$R_e = 101\,k\Omega$, $X_m = 1.6\,k\Omega$

(all seen from the 2400 side)

Current drawn from the mains under open-ckt condition.

Sol: Ref. all to LV side



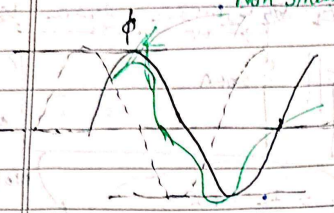
$R_{e(LV)} = \frac{10 \times 1000}{10^2} = 100\,\Omega$

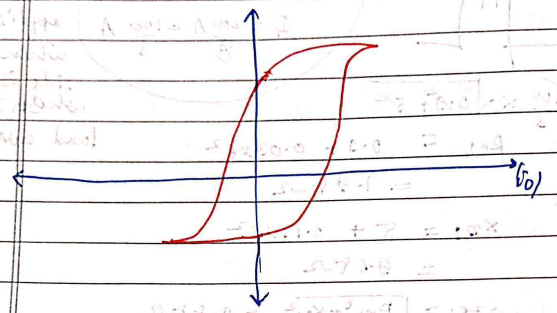
$X_{m(LV)} = \frac{1.6 \times 10^3}{10^2} = 16\,\Omega$

Friday

Classwork

Non sinusoidal peaky current @ no load





⇒ For establishing sinusoidal flux in core at no load the current (i_0) required is a non-sinusoidal peaky current.

@ FL condition

$\bar{I}_1 = \bar{I}_0 + \bar{I}_2$

$\bar{I}_0$: peaky

$\bar{I}_2$: load equivalent current

$\bar{I}_0$ → peaky current but very small

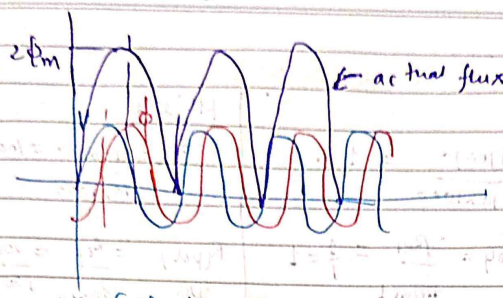
$\bar{I}_2$ → sinusoidal at 50 Hz

$\Rightarrow \bar{I}_1 \approx \bar{I}_2$

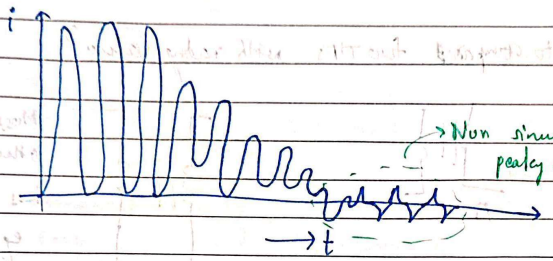
In rush current: die down after 3 cycles.

$\frac{V_1^2}{R_1} = \frac{V_2^2}{R_2}$

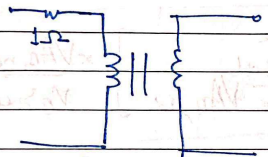
$R_2 = 100\ \Omega$



Current profile



* 100 VA



LV side	HV side
$R_{e1} = 1\ \Omega$	$R_{e2} = 100\ \Omega$

Base quantity : Z_{Base}

$$R_{per} = \frac{R_e}{Z_{Base}}$$

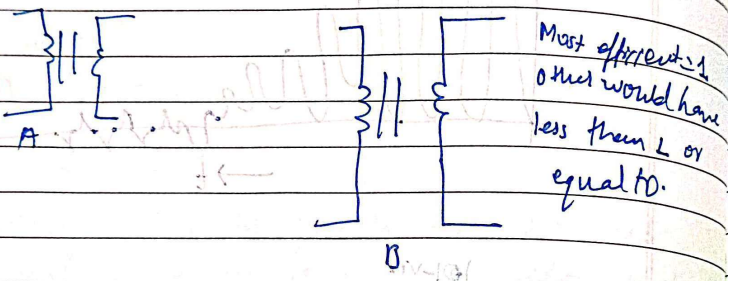
Per Unit quantity = $\frac{\text{Abs value of any quantity}}{\text{Base value of the same quantity.}}$

Now, $Z_{Base} = \frac{V_{Base}^2}{I_{Base} \cdot V_{Base}} = \frac{V_{Base}^2}{|VI|_{Base}} = \frac{V_{Base}^2}{S_{base}}$

So, here ;

LV side	HV side
$Z_{B_1} = \frac{(100)^2}{10 \times 1000} = 1\ \Omega$	$Z_{B_2} = \frac{(1000)^2}{10 \times 1000} = 100$
$R(pu) = \frac{R_{e1}}{Z_{Base\,1}} = \frac{1}{1} = 1$	$R(pu) = \frac{R_{e1}}{Z_{Base\,2}} = \frac{100}{100} = 1$

Q How to compared two TFs with rating values?



NOTE:

$$Z_{per,new} = \left[Z_{per,old} \times \frac{V_{B,old}^2}{VA_{B,old}} \right] \times \frac{VA_{B,new}}{V_{B,new}^2}$$

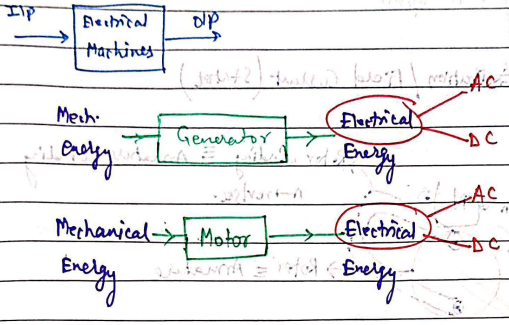
$i(\vec{v}\times\vec{B})$ $(\vec{v}\times\vec{B})\cdot\vec{l}$

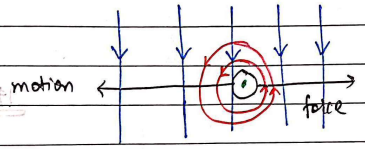
Monday

D.C. Machines:

Electro-mechanical Systems
(Machines)

Generators — Motors

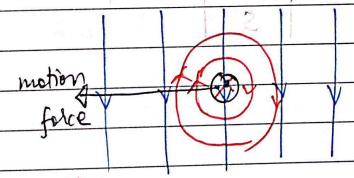




Law of Induction
(Generators)

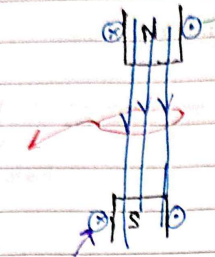
⊙ Current
⊙ Induced emf

NOTE: For energy conversion, there should be some agitation (oppose)



Law of Interaction
(Motors)

⊗ Current
⊙ emf

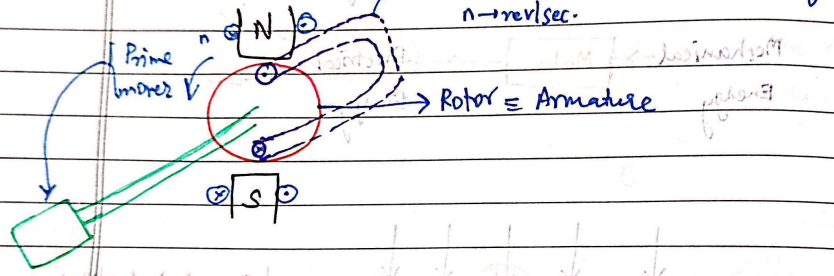


Why to use electromagnets?
⇒ To control $\vec{B}$, which is not possible for permanent magnets.

Fringing can be minimized by having low reluctance material b/w poles.

2-Pole System

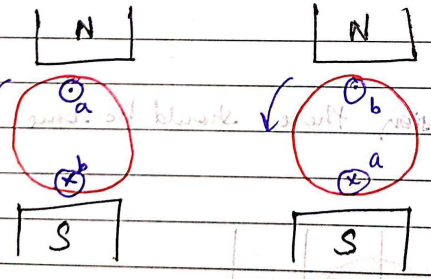
Excitation / Field Current (Stator)

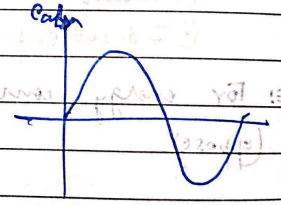


Tuesday

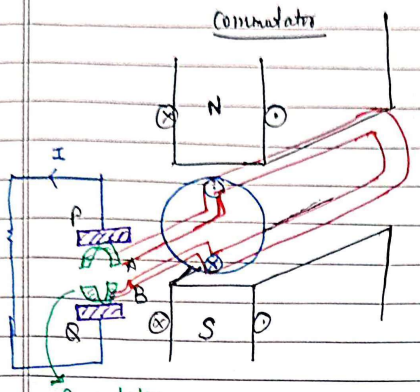
14 to 23

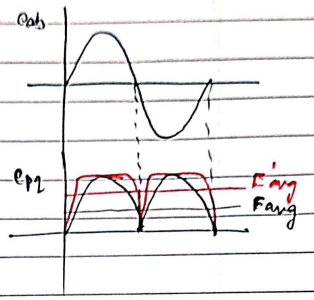
2-pole system



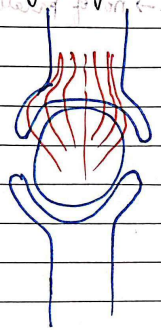


Rectifier





To get Eavg, slight modification is required !



Now, $e_{pg} = \frac{d\phi}{dt}$

$E_{coil} = \frac{\phi p}{(1/n)} = \phi . n . p$ volts

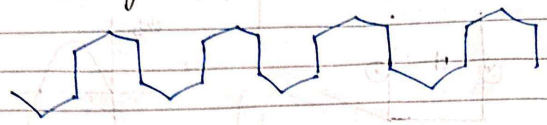
$E/coil = \frac{\phi \cdot \omega_m p}{2\pi}$

$$E = \frac{\phi . p . \omega_m \times Z}{2\pi}$$

$\phi \rightarrow$ flux/pole
$p \rightarrow$ no of poles
$n \rightarrow$ rev/sec
$1 \rightarrow 1/n$
$Z \rightarrow$ Total no. of conductors

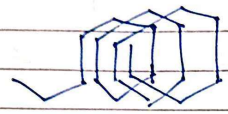
Wave winding → preferred for high V low I

No. of ||el paths
↑
L = P
↓
No. of poles



Lap windings → preferred for low V high I

L = P



$E = \left(\frac{PZ}{2\pi A}\right) \phi \omega_m$

A → no. of parallel paths

avg. value of total emf induced in Z no. of conductor

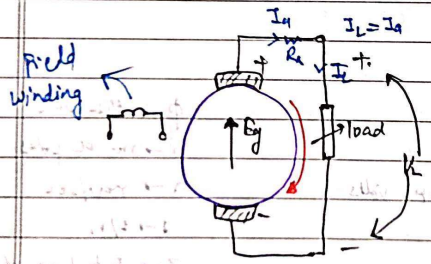
$E_g = K_e \phi \omega_m$; volt

emf constant

→ flux/pole ⇒ controlled by excitation/field current

emf const.

$K_e = \frac{PZ}{2\pi A}$



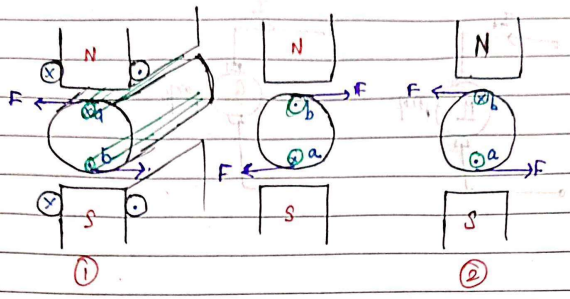
$E_g = V_t + R_a I_a$.

$R_a \rightarrow$ armature winding resistance

SEDCG: Field winding external to the machine

↓

Separately Excited DC Generator



⇒ We can go from ① to ② ; if same commutators are used.

∴ Commutator → DC/AC converter
→ DC/AC inverter

Now, $f_{coil} = B I_c l$; Newtons.

$= \frac{\phi_p}{A_c} \times I_c \times l$

$T_e = \gamma \times F_t = \left[\frac{\phi_p}{2\pi r l} \times I_a \times l \times \frac{z}{A}\right] \times r$

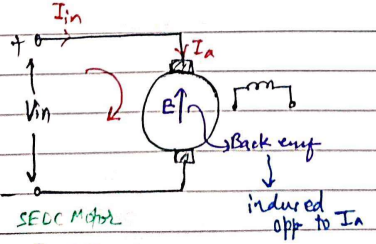
$K_e = K_t$ → Speed is in rad/sec.

$T_e = \frac{\phi_p}{2\pi r l} \times l \times r \times \frac{z}{A} \cdot I_a$

$\Rightarrow T_e = \left(\frac{pz}{2\pi A}\right) \cdot \phi \cdot I_a$

$\Rightarrow \boxed{T_e = K_t \phi I_a}$ N-m

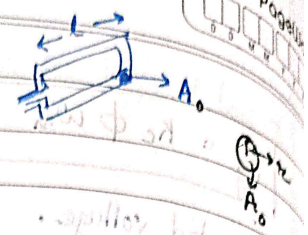
Torque constant



$V_{in} = E_b + R_a I_a$

$E_b = K_e \phi$

Length of Rotor = l
coil



$F_{coil} = B I_c l$ Newtons

$= \frac{\phi_p}{A_0} \times I_c \times l$

I_c : coil current

A_0 → cross section area of rotor coil

$T_e = F_t \times r = \left[\frac{\phi_p}{(2\pi \times r \times l)} \times I_a \times l \times \frac{z}{A}\right] \times r$

r → radius of rotor coil

I_a → Total armature current

→ Multiple windings in many ckts

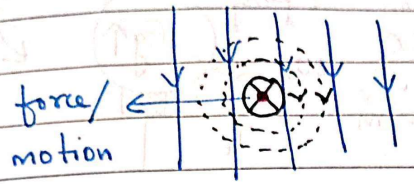
$T_e = \frac{\phi \times p}{2\pi r l} \times r \times \frac{z}{A} \times I_a$

$\boxed{T_e = K_t \phi I_a}$ N-m

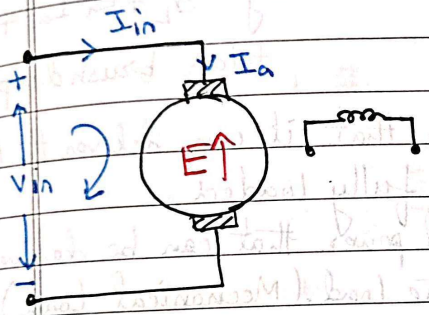
Torque constant

$K_e = K_t$ → Speed is rad/sec

Induced EMF generated which opposes dir of flow of current

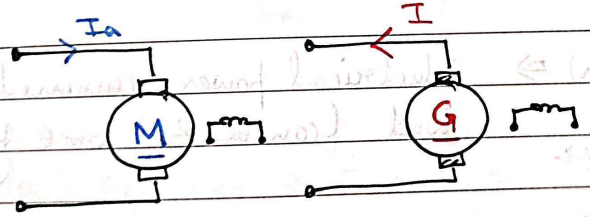


O x current
• emf



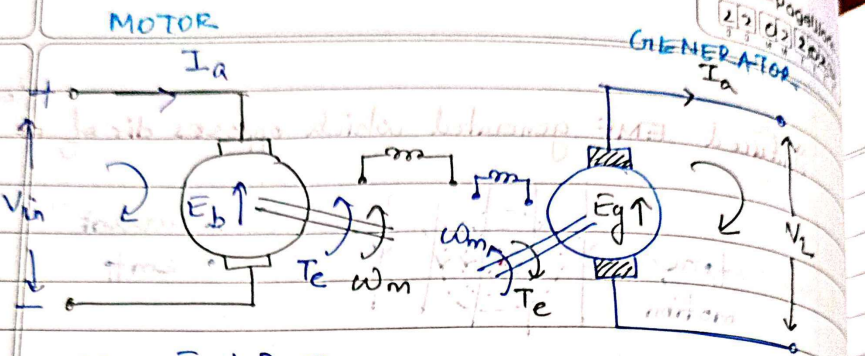
$$V_{in} = E_b + R_a I_a$$

Back emf



Draws current Supplies Current

Same machine can be made to work as M or G
Depends on dir of flow of current.



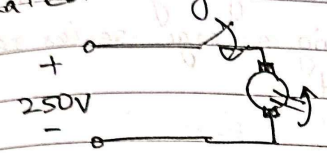
$V_{in} = E_b + R_a I_a$ + total brush drop

$E_g = V_L + R_a I_a$ + total brush drop

Motor Rated : 5kW ⇒ Power that it can deliver to load
↓ Mechanical Power when fully loaded
Power Consumed by motor > 5kW i.e. Rated power that can be delivered to the load. (Mechanical load)

Generator Rated 5kW ⇒ Electrical power consumed by load (can be R, L or C type)
↑ Output Power

10KW DC motor has armature resistance of 0.5Ω.
Rated voltage 250V



Current drawn from mains when switch is closed (Ignore brush drop)

Speed of rotor immediately after switch closed = 0

$V_{in} = E_b + R_a I_a \Rightarrow I_a = \frac{V - E_b}{R_a}$

$\Rightarrow I_a = \frac{V - K_e \phi \omega}{R_a}$ $\quad \omega = 0$ initially

$\Rightarrow I_a = \frac{V}{R_a} = \frac{250}{0.5} A = 500 A$

10KW Power Output drawn at 250V.
100% Efficient $\Rightarrow$ I = 40A. = full load/rated current

In comparison to I, I_a is huge.

∴ need a mechanism to apply variable voltage so that initially v low then gradually ↑.

OR

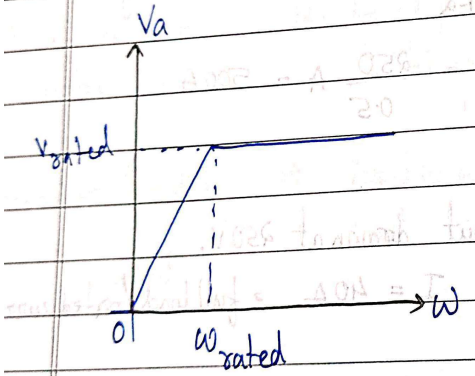
Have a resistance R_{st} in the beginning to have increased starting resistance. Keeping a large series resistance with the armature circuit.

$E_b = V - R_a I_a = K_e \phi \omega_m$.

$\omega_m \propto \frac{(V - R_a I_a)}{\phi} \propto \frac{V_a}{\phi}$

V_a → Armature voltage control.
↓
Using R_{st} to vary R_a.

ϕ → Field flux control.



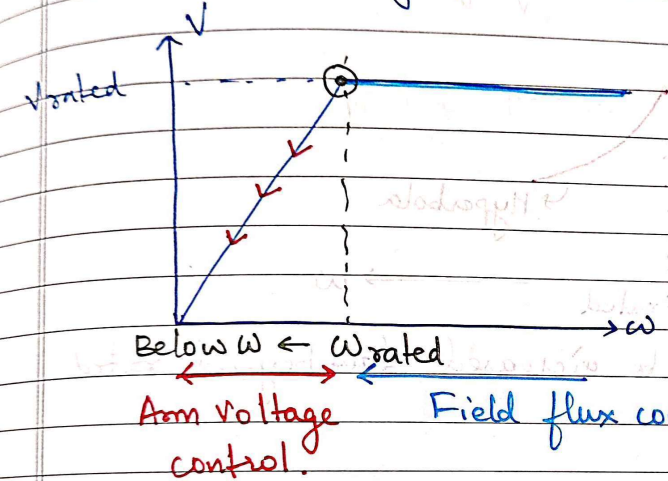
① ϕ → const V_a → varies. ⇒ ~~Can ↓ Vrated~~
Can ↓ V after V_{rated} reached. to ↓ ω to values less than ω_{rated}.

∴ Arm voltage control to achieve ~~speeds~~ below rated speeds.

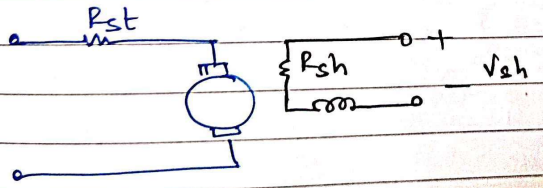
(5) $\phi \rightarrow$ vary $V_a \rightarrow$ const.

To achieve speeds above ω_{rated} ϕ is ↓.

{If V_a ↑ will reach saturation ∵ being operated very close to knee ~~voltage~~ point}.



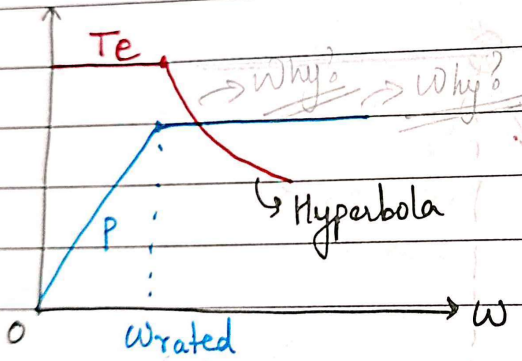
Field flux control ⇒ Depends on field circuit.



When beginning circuit: Rst high, Rsh low.

↓ To protect armature. ↓ To protect shaft

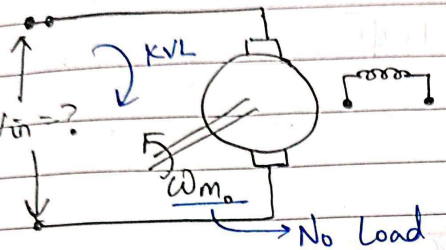
R_{sh} present in field winding, if R_{sh} high ⇒ i small, ϕ will be small ⇒ ω_m high ⇒ Shaft will ~~experience~~ experience shock loading. ∴ Should be kept min^m. at the ~~beg~~ starting of switch.



Power cannot be increased ~~too~~ beyond rated

24/02/2023

@ 40A → $\omega_m = 150$ rad/sec.



$R_a = 0.5\ \Omega$

$K_e\phi = 1.2\ V/(rad/s)$

$V = E_b + R_a I_a = (K_e\phi)\omega_m + R_a I_a$.

$V = (1.2 \times 150) + (0.5 \times 40) \Rightarrow \boxed{V_{in} = 200V}$

$E_b = V - R_a I_a \Rightarrow \omega_m = \dfrac{V}{K_e\phi} - \dfrac{R_a I_a}{(K_e\phi)^{Eq}}$

At no load, $I_a = 0$.

$\Rightarrow \omega_{m_0}$ (no load speed) $= \dfrac{V}{K_e\phi}$

$\Rightarrow \omega_{m_0} = \dfrac{200}{1.2} \simeq 166$ rad/sec.

N_{m_0} (in RPM) = 1042 RPM

$$\omega_m = \frac{V}{K_e\phi} - \frac{R_a I_a}{K_e\phi}$$

$T_e = (K_e\phi)\, I_a$.

$$\Rightarrow \omega_m = \frac{V}{K_e\phi} - \frac{R_a T_e}{(K_e\phi)^2}$$

T_L = Overcome losses (Dead load of machine i.e. stray torque) + Torque requirement of load

$$T_e = T_L + J\frac{d\omega_m}{dt}$$

T_e: EM torque that machine should actually develop.

$J\frac{d\omega_m}{dt}$: Dynamic Component to take inertia into account. ∴ May be +ve or −ve.

In steady state: $\boxed{T_e = T_L}$